

A Panel Analysis of Inequality and Homelessness

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Summary

This paper examines the relationship between income inequality and homelessness across U.S. states from 2010–2023 using a fixed-effects panel regression framework. Contrary to conventional expectations, the results indicate that higher income inequality is associated with lower homelessness rates after controlling for state-level economic and demographic factors. This finding suggests that the relationship between inequality and housing insecurity may be more complex than commonly assumed, potentially reflecting differences in housing markets, migration patterns, or measurement limitations across states. These results highlight the importance of careful empirical analysis when evaluating structural drivers of homelessness and suggest avenues for further research into the mechanisms underlying this relationship.

Introduction

Income inequality impacts everyone but for those at the bottom the issue becomes increasingly severe. Homelessness is a multifaceted, complex issue with no simple explanation on how to solve it. It reflects a crisis from mounting economic pressures that disproportionately impact society's most vulnerable. In 2014, taxpayers paid about \$35,500 per homeless person per year (National Alliance to End Homelessness, 2017), and despite these efforts homelessness remains a persistent problem 10 years later. While various factors like mental health, and substance abuse contribute to homelessness, income inequality is often overlooked (Quigley et al., 2001). The widening economic gap between the rich and the poor not only exacerbates financial strain, but also restricts access to essential resources like housing, pushing people into homelessness.

This paper attempts to explain the connection between income inequality and homelessness in the US, exploring the extent to which economic disparity influences homelessness rates. Using state-level panel data from 2010-2023 and a fixed effects regression model, I find that higher income inequality is significantly associated with lower levels of homelessness, contrary to most existing literature. After controlling for unemployment and housing stock per 100,000 residents, the negative relationship of inequality and homelessness

rates remains significant. These results also show that higher unemployment is significantly positively correlated to homelessness rates, with no significant relationship with housing stock.

As wealth increasingly concentrates at the top, those at the bottom are left with increasing financial burdens, less affordable housing options, and fewer economic opportunities. By exploring the relationship between economic disparity and homelessness, this study aims to illuminate how economic inequality fuels the growing crisis of homelessness.

Wealth accumulation has tilted dramatically in favor of higher earners since the 1980s. This prosperity at the top of the ladder is driven by the idea that wealth would “trickle down” from the top to the bottom of the ladder (Horowitz et al., 2020). For many, this has been an empty promise. Instead, stagnant wages, soaring cost of living, and increasingly expensive housing has made financial stability difficult for those at the lower end of the income distribution (Quigley et al., 2001). Housing, which some consider a basic right, has become unattainable for many people.

Income inequality does not only reflect a widening gap between high and low wage earners, it actively contributes to homelessness by concentrating economic resources and power into the hands of a few. This concentration diverts resources away from lower-income individuals who may rely on public assistance and affordable housing to maintain stability. Without effective, evidence-based policy intervention, homelessness will likely continue to impose substantial human and fiscal costs. Understanding the connection between economic inequality and homelessness is crucial for policymakers to make informed decisions and effective solutions.

Literature Review

In order to examine the current state of homelessness in the US, first we must understand how it developed. In the early 20th century, homelessness was distinct from what we see today. The common image of the era is of transient workers, depicted carrying the belongings in a bandana, hopping on trains in search of employment. In contrast, current homelessness is more closely associated with mental health and substance abuse issues (Byrne et al., n.d.). The rise of modern homelessness can be traced back to the 1980s. This period saw several socio-economic shifts. Gentrification of urban centers, defunding mental health services, high unemployment,

and significant budget cuts to the Department of Housing and Urban Development (HUD) (National Academies of Sciences, Engineering, and Medicine National Academies of Sciences, Engineering, and Medicine, 2018). Since then, services supporting health, housing, and economic security have declined, while wealth and earning disparities have ballooned, disproportionately benefitting higher-income individuals. For example, from 2018-2022, California alone has spent over \$17.5 billion to alleviate homelessness rates. In that timeframe the amount of homeless people has actually increased (Watt, 2023). These trends indicate a need for policy adjustments informed by a deeper understanding of homelessness and its root causes.

Homelessness is not a new phenomenon, and numerous studies have attempted to increase our understanding of its core factors. One study, *A Rising Tide Drowns Unstable Boats: How Inequality Creates Homelessness*, (Byrne et al., n.d.) investigates how income inequality acts as a driver of homelessness and the mechanisms behind the relationship. The authors question if an income or price channel drives homelessness. The income channel suggests that as income growth disproportionately favors higher earners, they increasingly dominate the housing market. They outbid and drive up housing costs, while lower earners have to allocate a larger share of their income towards housing. The price channel is where high income earners elevate the total demand in the housing market. This leads to rising costs which eventually some individuals can no longer afford. This study utilizes an OLS model and an instrumental variable (IV) approach to conduct the analysis. The IV model is employed by the researchers in order to establish the direction of causality, ensuring that income inequality impacts homelessness and not the other way around. The researchers use “national changes in the income distribution as an instrument for variation in a community’s actual income distribution”(Byrne et al., n.d.). They assume that national changes are an exogenous source of variation at a community level. One concern of this study is their paper does not mention their inclusion or exclusion criteria. These criteria are an important aspect of conducting IV model regression. This study finds a causal relationship between income inequality and homelessness, primarily driven by the income channel, high earners crowding out the housing market.

Data & Empirical Methodology

This study examines the relationship between income inequality and homelessness in the United States using data from four primary sources: U.S. Department of Housing and Urban Development (HUD), American Community Survey (ACS), Rhode Island Department of Labor and Training (DLT) and the Census Bureau. The dataset covers the years 2010-2023, as this is the common period available across all four sources.

1. **Income Inequality Data (ACS):** The income inequality dataset measured by the ACS using the Gini Coefficient. The datasets have 2 variables, state and gini coefficient, with 52 observations, the 50 US states, Washington D.C. and Puerto Rico. The Gini Coefficient shows income inequality as a number from 0-1. Perfectly equal societies have a coefficient of 0, increasing to 1 as all wealth becomes more concentrated in the hands of one person. The ACS reports the Gini Index by state from the years 2010-2023. The data was cleaned by removing headers, extraneous “Estimate” rows, and aligning state-level observations across the years. The final dataset had three columns: State, Year, and Gini Coefficient.
2. **Homelessness Data (HUD):** The homelessness dataset reports homelessness from continuums of care (CoCs). The raw dataset has a differing number of observations through the years due to CoC changes, but after aggregating to the state level we will only have 56 observations, one for each state and the US territories. It shows which state the CoC is in, total counts of homeless per CoC, and homelessness by gender, age, and/or race. The PIT count provides a point-in-time estimate of people experiencing homelessness on a single night in January, including sheltered and unsheltered populations, though measurement limitations and undercounting remain important concerns. The data is reported for the years 2007-2023. For this analysis the homelessness totals reported by the CoC will be aggregated to the state level. This data was cleaned by removing unneeded data, variables breaking down the overall homeless into race, gender, and age groups, and then combining the sheets (one sheet is one year of observations) into one dataset. It has columns for state, year, and overall homeless.
3. **Unemployment Data (DLT):** This dataset shows unemployment rates for all US states and Washington D.C. for 2000-2024. This dataset also shows the rank of each state, with rank

1 having the least unemployment, and rank 51 being the highest unemployment. To clean this data, the rank was removed and the unemployment rates associated with each state were moved to the final dataset.

4. Population and Housing Stock Data (Census): The State population dataset has data going back to the 1900s, but for this analysis we will only be using data from 2010-2023 as that is the limit of our income inequality data. The dataset has 2 variables, state and population, with 52 observations. The housing stock data shows total housing units per state over the years 2010-2024. The population data will be used to get a homelessness per 100,000 variable and housing per 100,000 variable. A per capita variable is necessary when comparing states as just comparing total numbers would create bias between states with varying populations. This data was cleaned by removing headers, regional totals, and creating a column for the years.

After cleaning and standardizing the three datasets, they were merged by state and year into a balanced panel dataset. The resulting dataset contained 52 states and territories observed over 14 years, 2010-2023, totalling 728 observations. This paper's focus is on the U.S. states so Washington D.C. and Puerto Rico were excluded in the final analysis, leaving us 50 entities, with 700 observations. The dataset enables panel data analysis techniques to examine the relationship between income inequality (Gini Coefficient) and homelessness per 100,000 people while controlling for time and state characteristics. The summary statistics for the Gini Coefficient looks like:

Figure 1:

Variable	Mean	Std. Dev.	Min	Max
Homeless per 100,000	154.31	97.7	21.24	554.86
Gini Coefficient	.4631	.0192	.4081	.5208
Housing per 100,000	44,158	3,338.87	34,938	55,865
Unemployment Rate (%)	5.45	2.24	1.8	13.5

Using an OLS model, this paper will attempt to estimate the impact of income inequality on the rate of homelessness. Our variable of interest is overall homeless, on which we will

regress the Gini Coefficient. Given we have a panel of state data, state and year fixed effects will be used to control for temporal differences and state-to-state differences in our estimate. The regression equation will look like so:

$$\text{Homeless per 100000}_{it} = B_0 + B_1(\text{Gini Coefficient})_{it} + \alpha_i + \gamma_t + E_{it}$$

Where:

- $\text{Homeless per 100000}_{it}$ is the number of homeless individuals per 100,000 people in state i at time t ,
- $\text{Gini Coefficient}_{it}$ is the measure of income inequality in state i at time t ,
- α_i represents the state fixed effects, capturing unobserved characteristics of each state,
- γ_t represents the year fixed effects, accounting for trends over time,
- E_{it} is the error term that captures unobserved variation.

By adding in the state and year fixed effects, we control for state-to-state differences, and year-to-year changes, removing bias from unobserved, time-invariant state-level characteristics.

Results

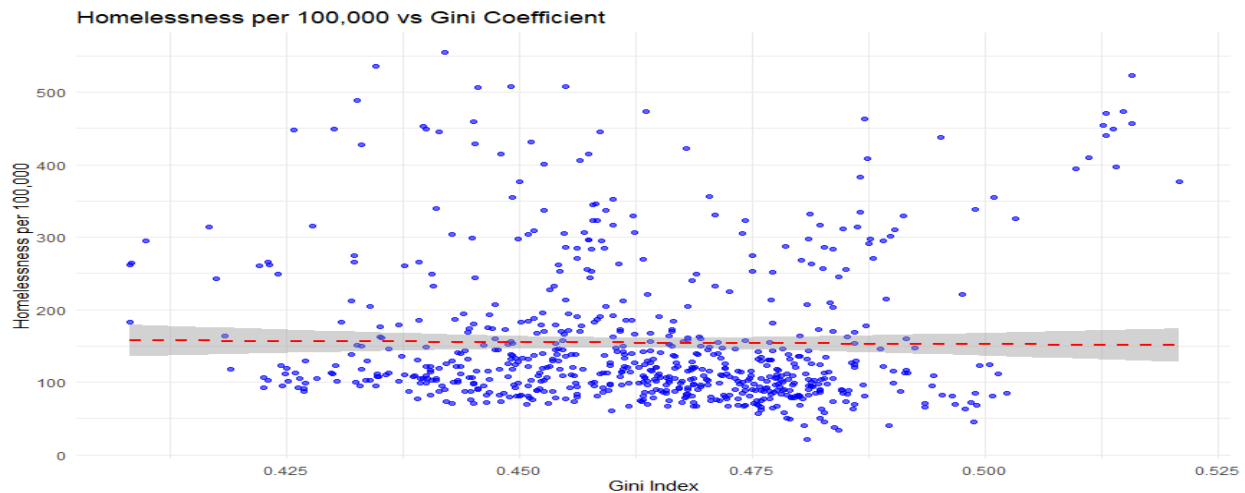
The results from the fixed effects model:

Figure 2:

Dependent variable:	
homeless_per_100000	
Gini_Coefficient	-1,385.762*** (221.372)
Observations	700
R2	0.057
Adjusted R2	-0.016
F Statistic	39.186*** (df = 1; 649)
Note: *p<0.1; **p<0.05; ***p<0.01	

Figure 3 below visualizes the estimated relationship between the Gini Coefficient and homelessness rate per 100,000 people. The trend line shows the predicted values from the fixed effects model, controlling for state and year. Each point represents a state-year observation.

Figure 3:



Visually, we can see that the regression line is nearly straight and with the grey area around the line not being uniform, we may have heteroskedasticity. This is further confirmed by a Breusch-Pagan test which shows:

Figure 4:

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studentized Breusch-Pagan test
data:  reg
BP = 5.781, df = 1, p-value = 0.0162
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Given our P-value is less than .05, we reject the null hypothesis that there is no heteroskedasticity at the 5% significance.

We do see a statistically significant coefficient for the Gini Coefficient at the 1% level, but it being negative is going against the direction we predicted it would. The -1,385.76 value implies that an increase in the Gini Coefficient by 1 unit leads to 1,385.76 fewer homeless people. Since the Gini Coefficient ranges from 0-1, we should analyze how a .01 increase, a realistic number, in the Gini Coefficient impacts homelessness. A .01 increase in the Gini Coefficient would lead to a 13.86 decrease in homeless individuals per 100,000 people in a state.

While the results are statistically significant, the R^2 value of .057 indicates that about 5.7% of the in-state variation of homelessness rates is explained by variation in the Gini Coefficient. The negative adjusted R^2 suggests that our model's predictions are worse than just

using the average of our dependent variable to predict the rate of homelessness. However our F-statistic of 39.186 is highly significant, signalling that at least one of the variables or controls are explaining the variation in homelessness.

The negative sign on the Gini Coefficient is unexpected, as most literature suggests that higher income inequality leads to higher levels of homelessness, not a reduction as our model suggests. This could be due to a multitude of reasons. Our model does not include variables such as housing costs, mental health support, or unemployment, which may impact homelessness regardless of inequality leading to omitted variable bias. Also, some states with higher income inequality may have stronger social safety nets that might mitigate the effect of inequality.

I attempted to fix the heteroskedasticity issue by introducing a log transformation on each of our variables. This resulted:

Figure 5:

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=====
                        Dependent variable:
                        -----
                        log(homeless_per_100000)
                        -----
log(Gini_Coefficient)      -0.485
                           (0.474)
                        -----
Observations                700
R2                          0.002
Adjusted R2                 -0.019
F Statistic                 1.045 (df = 1; 685)
=====
Note:                       *p<0.1; **p<0.05; ***p<0.01
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This log transformed regression does not show statistical significance at any relevant level, low explanatory power with a .002 R^2 , and the F-statistic of 1.045 further suggests that this model does not show us anything new. The Breusch Pagan Test shows:

Figure 6:

```
studentized Breusch-Pagan test
data:  reg2
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BP = 24.149, df = 1, p-value = 8.917e-07

The reason for running the log transformed regression was to try and eliminate heteroskedasticity, but the BP test gives an even lower p-value than the initial regression showing that heteroskedasticity is still very likely present.

After these models showed a negative and statistically significant coefficient on the Gini Coefficient, I introduced two control variables, housing units per 100,000 residents and unemployment rate. This regression resulted:

Figure 7:

Dependent variable:	
homeless_per_100000	
Gini_Coefficient	-916.138*** (256.526)
housing_per_100000	-0.002 (0.003)
unemp_rate	3.001*** (0.857)
Observations	700
R2	0.076
Adjusted R2	0.002
F Statistic	17.713*** (df = 3; 647)
Note: *p<0.1; **p<0.05; ***p<0.01	

The coefficient on the Gini Coefficient remains negative and significant, even when accounting for housing stock and unemployment rates. This suggests that states with higher income inequality still show lower rates of homelessness according to this model, a result that contradicts the existing literature. The coefficient on unemployment suggests that a 1% increase in unemployment leads to 3 additional homeless people per 100,000 residents. The housing stock variable is not significant, which suggests that housing stock alone does not explain variation in

homelessness across states. The model does have a slightly higher R^2 and we finally see a positive adjusted R^2 which means this model may be better than using the mean of the dependent variable to predict homelessness. This shows that for future models, more controls will probably bring us closer to the true impact of income inequality on homelessness.

Conclusion

This paper investigated the relationship between income inequality and homelessness in the United States using state-level panel data from 2010-2023. While the literature strongly suggests that higher income inequality leads to higher homelessness rates, my findings show a statistically significant negative relationship between the Gini Coefficient and homelessness rates. Even after controlling for unemployment rates and housing stock per 100,000 residents, the negative relationship persists. This unexpected result most likely reflects omitted variable bias, or policy differences in counties or cities in each state.

A few implications arose from this analysis. First, the results underscore the importance of meaningful controls, such as the unemployment rate, which shows a significant and positive relationship with homelessness. This highlights the role of economic security when analyzing homelessness. Second, the insignificant effect of housing stock suggests that merely looking at the amount of housing is not a good indicator. Affordable housing stock or housing policies may be a better indicator of homelessness. Also, the literature suggests that low earners are crowded out from the housing market, so looking at median rent-to-income might also be a direction to look into.

These findings emphasize the complexity of homelessness. A simple model with only a few controls, does not capture the true impact of income inequality on homelessness. Looking more local, at the county or city, would most likely help in isolating the effect of inequality as there is much variation within states. Policymakers should approach the multifaceted issue of homelessness with a toolkit of economic support, affordable housing, and mental health services.

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